

Assignment 3: Solutions

Throughout, G denotes a simple graph with $n \geq 3$ vertices.

1. Hamiltonian condition via Ore's theorem and its tightness

(i) If $m = \frac{(n-1)(n-2)}{2} + 2$, then G has a Hamiltonian cycle.

Strategy. We prove by contradiction that G satisfies Ore's condition:

$$\text{for every nonadjacent } v, w, \quad \deg(v) + \deg(w) \geq n.$$

By the version of Ore's theorem quoted in class, this implies that G is Hamiltonian. $\square$

Proof. Assume there exist nonadjacent $v, w \in V(G)$ with

$$\deg(v) + \deg(w) \leq n - 1.$$

Upper bound the number of edges $m = |E(G)|$ by separating edges incident to $\{v, w\}$ from all others:

$$m \leq \binom{n-2}{2} + (\deg(v) + \deg(w)).$$

Indeed, the $\binom{n-2}{2}$ term counts all edges entirely inside $V(G) \setminus \{v, w\}$, and then we add at most $\deg(v) + \deg(w)$ edges that use v or w ; the edge vw does not exist.

Using $\deg(v) + \deg(w) \leq n - 1$,

$$m \leq \binom{n-2}{2} + (n-1) = \frac{(n-2)(n-3)}{2} + (n-1) = \frac{n^2 - 3n + 4}{2}.$$

But the given hypothesis is

$$m \geq \frac{(n-1)(n-2)}{2} + 2 = \frac{n^2 - 3n + 2}{2} + 2 = \frac{n^2 - 3n + 6}{2},$$

which exceeds the previous upper bound by 1:

$$\frac{n^2 - 3n + 6}{2} - \frac{n^2 - 3n + 4}{2} = 1.$$

This contradiction shows no such nonadjacent pair can violate Ore's condition; hence $\deg(v) + \deg(w) \geq n$ for all nonadjacent v, w . By Theorem 1 (Ore), G has a Hamiltonian cycle. $\square$

(ii) Replacing $\deg(v) + \deg(w) \geq n$ by $\deg(v) + \deg(w) \geq n - 1$ is false.

Proof. Consider G obtained from the complete graph K_{n-1} by attaching one new vertex x with a single edge to a vertex of K_{n-1} (i.e. a pendant vertex). Then G is not Hamiltonian because any Hamiltonian cycle would give $\deg(x) \geq 2$, yet $\deg(x) = 1$.

We verify the weakened condition: the only nonadjacent pairs involve x and a vertex y of K_{n-1} that is *not* the neighbor of x . For such a pair,

$$\deg(x) + \deg(y) = 1 + (n - 2) = n - 1.$$

There are no other nonadjacent pairs (the K_{n-1} part is complete). Hence the condition $\deg(v) + \deg(w) \geq n - 1$ holds for every nonadjacent pair, but G is not Hamiltonian. Therefore the theorem fails with $n - 1$. $\square$

2. An edge is a bridge iff it lies in no cycle

To Prove: *An edge e of G is a bridge (its removal increases the number of connected components) if and only if e belongs to no cycle of G .*

Proof. ($\Rightarrow$) Suppose $e = uv$ is a bridge. If e were on a cycle, then that cycle would contain a u - v path not using e . After removing e , u and v would remain connected via that path, contradicting that e is a bridge. Hence a bridge cannot lie on any cycle.

($\Leftarrow$) Conversely, suppose $e = uv$ belongs to no cycle. Consider $G - e$. If u and v were still connected in $G - e$, there would exist a u - v path P not using e . Then $P \cup \{e\}$ would form a cycle containing e , a contradiction. Therefore removing e disconnects u and v , increasing the number of components; i.e. e is a bridge. $\square$

3. A walk implies a path between the same vertices

To Prove: *If there is a (finite) walk from u to v in G , then there exists a path from u to v .*

Proof. Let $W = (u = x_0, x_1, \dots, x_k = v)$ be a u - v walk (vertices may repeat). If all vertices are distinct, W is already a path. Otherwise, there exist indices $0 \leq i < j \leq k$ with $x_i = x_j$. Remove the closed subwalk $(x_i, x_{i+1}, \dots, x_j)$ to obtain a shorter u - v walk. Repeating this cycle-erasure procedure finitely many times yields a u - v walk with no repeated vertices, i.e. a path. $\square$

4. Every closed odd walk contains an odd cycle

To Prove: *Every closed walk of odd length in G contains a cycle of odd length.*

Proof. Let $W = (x_0, x_1, \dots, x_{2\ell}, x_{2\ell+1} = x_0)$ be a closed walk of odd length $2\ell + 1$. If W has no repeated vertices other than x_0 at the ends, then W is itself an odd cycle and we are done.

Otherwise, let x_i be the first vertex that repeats in W , i.e. there exist $0 \leq i < j \leq 2\ell + 1$ with $x_i = x_j$, and the segment $(x_i, \dots, x_j)$ contains no internal repetitions. Then $(x_i, \dots, x_j)$ is a closed walk. If $j - i$ is odd, this subwalk is an odd cycle (no vertex repeats except its ends, by choice of j minimal), and we are done.

If $j - i$ is even, remove this even closed subwalk from W to form a shorter closed walk W' . The parity of the total length flips modulo 2 only when removing odd-length pieces; since we removed an even-length piece, W' remains odd and strictly shorter. Iterating this process must eventually terminate (length decreases), and the final odd closed walk has no internal repetitions, hence is an odd cycle. $\square$

5. Minimum degree $\geq \frac{n-1}{2}$ implies connectedness

To Prove: *If every vertex of G has degree at least $\frac{n-1}{2}$, then G is connected.*

Proof. Assume, for contradiction, that G is disconnected. Let C be a connected component with $t = |V(C)|$ vertices, $1 \leq t \leq n-1$. Any vertex in C has no neighbors outside C , so its degree is at most $t-1$. By the hypothesis,

$$t-1 \geq \frac{n-1}{2} \quad \Rightarrow \quad t \geq \frac{n+1}{2}.$$

Apply the same argument to any other component C' , which must also satisfy $|V(C')| \geq \frac{n+1}{2}$. But then

$$|V(C)| + |V(C')| \geq \frac{n+1}{2} + \frac{n+1}{2} = n+1,$$

impossible since G has only n vertices. Therefore G must be connected. $\square$

6. No closed knight's tour on an odd $n \times n$ board

Fix $n \in \mathbb{N}$ and let $C = \{1, \dots, n\}^2$ be the set of board squares with coordinates (i, j) . The *knight graph* K_n has vertex set $V(K_n) = C$ and an edge between (i, j) and (i', j') iff $\{|i - i'|, |j - j'|\} = \{1, 2\}$ (i.e. a legal knight move). A *closed knight's tour* is a Hamiltonian cycle in K_n .

Lemma 1 (Bipartiteness). *Color each square (i, j) black if $i + j$ is even and white if $i + j$ is odd. Then every edge of K_n joins a black square to a white square. Hence K_n is bipartite with parts*

$$B = \{(i, j) : i + j \text{ even}\}, \quad W = \{(i, j) : i + j \text{ odd}\}.$$

Proof. If (i, j) and (i', j') are joined by a knight move, then $\{|i - i'|, |j - j'|\} = \{1, 2\}$, so $i - i'$ and $j - j'$ have opposite parity. Therefore $(i + j) - (i' + j')$ is odd; equivalently $i + j$ and $i' + j'$ have opposite parity. Thus every edge crosses B and W . $\square$

Lemma 2. *If n is odd, then $|B| = \frac{n^2 + 1}{2}$ and $|W| = \frac{n^2 - 1}{2}$.*

Proof. On each row (fixed i), the colors alternate and, because n is odd, there is one more square of the row's starting color. Rows with i odd start with black at $(i, 1)$ (since $i + 1$ is even), while rows with i even start with white. Since there are $(n + 1)/2$ odd rows and $(n - 1)/2$ even rows,

$$|B| = \frac{n+1}{2} \cdot \frac{n+1}{2} + \frac{n-1}{2} \cdot \frac{n-1}{2} = \frac{(n+1)^2 + (n-1)^2}{4} = \frac{n^2 + 1}{2}.$$

Then $|W| = n^2 - |B| = \frac{n^2 - 1}{2}$. $\square$

Argument: *If n is odd, the knight graph K_n has no Hamiltonian cycle. Equivalently, a closed knight's tour on an odd $n \times n$ board does not exist.*

Proof. **(A) Cycle-parity argument.** By Lemma 1, K_n is bipartite. Every cycle in a bipartite graph has even length (its vertices alternate between the two parts and must return to the start after an even number of steps). A Hamiltonian cycle would have length $|V(K_n)| = n^2$, which is odd for odd n —a contradiction.

(B) Balance-of-parts argument. In any bipartite graph, a Hamiltonian cycle must alternate between the two parts and therefore uses the same number of vertices from each part. By Lemma 2, when n is odd we have $|B| \neq |W|$, so such an alternating cycle cannot cover all vertices. Hence no Hamiltonian cycle exists. $\square$

7. Spanning trees and bridges

(a) Every connected graph has a spanning tree.

Proof. Let G be connected. Consider the set $\mathcal{H}$ of all connected spanning subgraphs of G . Order $\mathcal{H}$ by inclusion on edge sets. Since $G \in \mathcal{H}$, the set is nonempty. By Zorn's Lemma on the *reverse* inclusion order, or more simply by finite descent, there exists a minimal connected spanning subgraph T with respect to the number of edges.

We claim T is acyclic. If T had a cycle, remove any edge e on that cycle; the resulting graph remains connected (because the rest of the cycle still connects its endpoints) and is still spanning, contradicting minimality. Hence T is connected and acyclic, i.e. a tree. Thus T is a spanning tree of G . $\square$

(b) A graph with more than one spanning tree.

Example. Take the triangle $G = K_3$ with vertices $\{a, b, c\}$. Any two edges form a spanning tree (a path on 3 vertices). The three choices $\{ab, bc\}$, $\{bc, ca\}$, $\{ca, ab\}$ are distinct spanning trees. Therefore a graph can have multiple spanning trees. $\square$

(c) Edges in all spanning trees are exactly the bridges.

Theorem: For an edge e of a connected graph G , the following are equivalent:

- (i) e is present in every spanning tree of G ;
- (ii) e is a bridge of G (i.e. $G - e$ is disconnected).

Proof. $(\Rightarrow)$ Suppose e lies in every spanning tree but e is *not* a bridge. Then $G - e$ is still connected. By part (a), $G - e$ has a spanning tree T' (which is also a spanning tree of G). But T' omits e , contradicting that e is in every spanning tree. Hence e must be a bridge.

$(\Leftarrow)$ Suppose e is a bridge; write $e = uv$. Then $G - e$ has (at least) two components, say with u in C_1 and v in C_2 . Any spanning tree T of G is connected and acyclic. If $e \notin T$, then T would be a connected spanning subgraph of $G - e$, impossible since $G - e$ is disconnected. Therefore every spanning tree must include e . $\square$

8. Walk counting via powers of the adjacency matrix

To Prove : For all integers $k \geq 1$, the (i, j) -entry of A^k equals the number of $v_i \rightarrow v_j$ walks of length k in G .

Proof. We proceed by induction on k .

Base $k = 1$. The (i, j) -entry of A is a_{ij} , which equals 1 if v_i and v_j are adjacent and 0 otherwise. This is exactly the number of length-1 walks from v_i to v_j .

Inductive step. Assume the claim holds for $k \geq 1$; i.e., $(A^k)_{i\ell}$ equals the number of $v_i \rightarrow v_\ell$ walks of length k . Consider $A^{k+1} = A^k A$. Then

$$(A^{k+1})_{ij} = \sum_{\ell=1}^n (A^k)_{i\ell} a_{\ell j}.$$

Each term counts the number of ways to take a length- k walk from v_i to an intermediate vertex v_ℓ (times an indicator $a_{\ell j}$, which is 1 precisely when there is an edge $v_\ell v_j$ to extend by one step). Thus the sum counts the number of concatenations of a length- k walk $v_i \rightsquigarrow v_\ell$ with an adjacent final step $v_\ell v_j$, i.e. the total number of length- $(k+1)$ walks from v_i to v_j .

By induction, the statement holds for all $k \in \mathbb{N}$. □

Hall's Theorem: Problems

1. Regular bipartite graphs have an X -perfect matching

To Prove: Let $G = (X \cup Y, E)$ be a d -regular bipartite graph with $d \geq 1$. Then there exists a matching that saturates X .

Proof. Fix any $S \subseteq X$. Every vertex of S has degree d , and all its incident edges go into $N(S) \subseteq Y$. Hence

$$e(S, N(S)) = \sum_{x \in S} \deg(x) = d|S|.$$

On the other hand,

$$e(S, N(S)) \leq \sum_{y \in N(S)} \deg(y) = d|N(S)|.$$

Therefore $d|S| \leq d|N(S)|$, so $|N(S)| \geq |S|$. By Hall's theorem, G has a matching saturating X . □

2. Every Latin rectangle can be completed to a Latin square

To Prove: Every $r \times n$ Latin rectangle over symbols $\{a_1, \dots, a_n\}$ with $r \leq n$ extends to an $n \times n$ Latin square.

Proof. Proceed inductively on the number t of filled rows. Assume rows $1, \dots, t$ ($t < n$) form a Latin rectangle. Construct a bipartite graph $H = (L \cup C, E)$ where $L = \{a_1, \dots, a_n\}$ and $C = \{1, \dots, n\}$, and place an edge (a_j, c) iff a_j does not appear in column c among the first t rows. A completion of row $t+1$ is equivalent to a perfect matching in H .

We verify Hall's condition. Let $S \subseteq L$. Write $N(S) \subseteq C$ for its neighbors. If $c \notin N(S)$, then no symbol of S is available for column c , so all t used entries in column c lie in $L \setminus S$. Counting (entry, symbol) pairs with symbols in $L \setminus S$ across all columns:

$$t(n - |S|) \geq \sum_{c \in C} \#\{\text{entries in column } c \text{ among first } t \text{ rows that lie in } L \setminus S\} \geq t \cdot |C \setminus N(S)|.$$

Thus $t(n - |S|) \geq t(n - |N(S)|)$, and hence $|N(S)| \geq |S|$. By Hall, H has a perfect matching; fill row $t+1$ accordingly. Iterate until $t = n$. □

Matchings and Turán Theorem: Problems

1. A maximal matching that is not maximum

Proposition 1. *In the path P_4 with vertices $a - b - c - d$, the matching $M = \{bc\}$ is maximal but not maximum.*

Proof. Any edge other than bc shares an endpoint with bc , so no further edge can be added to M ; thus M is maximal. Yet $\{ab, cd\}$ is a matching of size $2 > |M|$, so M is not maximum. $\square$

2. A graph on n vertices whose maximum matching has size 1

Proposition 2. *The star $K_{1,n-1}$ has matching number 1.*

Proof. All edges are incident to the center; hence any two edges intersect. Therefore every matching has at most one edge, and choosing any leaf edge attains size 1. $\square$

3. Augmenting-path lemma toward Hall's theorem

To Prove: *Let $G = (X \cup Y, E)$ be bipartite and M a maximum matching with some $x \in X$ unmatched. Let S be the set of vertices in $(X \setminus \{x\}) \cup Y$ reachable from x by alternating paths (starting with a non- M edge), and set $S' = (S \cap X) \cup \{x\}$. Then $|N(S')| < |S'|$.*

Proof. **Claim 1:** $N(S') = S \cap Y$. Indeed, if $y \in S \cap Y$, then some alternating path from x ends at y with a non- M edge, so y has a neighbor in S' . Conversely, if $y \in N(S')$, it is adjacent to some $x' \in S'$, and by appending this non- M edge to the alternating path to x' we see $y \in S$; hence $y \in S \cap Y$.

Claim 2: Every $y \in S \cap Y$ is M -matched and its mate lies in $S \cap X$. If some $y \in S \cap Y$ were unmatched, the alternating path to y would be augmenting, contradicting maximality of M . If y is matched to x_y , then the matching edge $x_y y$ extends the alternating path to y , keeping alternation, so $x_y \in S \cap X$.

Thus the matching M induces a bijection between $S \cap Y$ and $(S \cap X)$, while x is an additional vertex in S' not matched by M . Therefore

$$|N(S')| = |S \cap Y| = |S \cap X| \leq |S \cap X| + 0 = |S'| - 1 < |S'|.$$

$\square$

4. Turán tightness for triangles

Proposition 3. *For even n , the complete bipartite graph $K_{n/2, n/2}$ has $n^2/4$ edges and contains no triangle; hence the $\lfloor n^2/4 \rfloor$ bound for triangle-free graphs is tight.*

Proof. $|E(K_{a,b})| = ab$, so with $a = b = n/2$ we have $|E| = n^2/4$. Bipartite graphs have no odd cycles, in particular no 3-cycles. Since Mantel's/Turán's bound for K_3 -free graphs is $\lfloor n^2/4 \rfloor$, this construction attains it for even n . $\square$